

# Development process of a wearable device for monitoring elderly people with dementia: State of development

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**Abstract.** The aging global population presents challenges in the management of neurodegenerative diseases such as dementia. This paper describes the development of a wearable system that aims to monitor and support elderly individuals with dementia. Key functional requirements were identified, leading to the creation of an initial functional prototype. The system integrates specific sensors such as BMI270, MAX30101, and SAM-M10Q, as well as the use of artificial intelligence to continuously and accurately monitor health indicators such as heart rate, blood oxygen levels, and daily activities. Focused on comfort and usability, the device is non-intrusive and suitable for continuous wear. Machine learning algorithms process the collected data to enable reliable activity recognition and health monitoring. Although user-based usability tests have not yet been performed, ergonomic and design studies have guided prototype development. The preliminary results of internal technical tests indicate stable sensor data acquisition and successful wireless communication using Bluetooth Low Energy (BLE). These results support the feasibility of the system and prepare the next development phase, which focuses on optimizing the hardware for a compact PCB (Printed Circuit Board) together with the redesign of the three-dimensional model, prioritizing its usability according to the ergonomic studies carried out.

**Keywords:** Wearable Devices · Dementia · Artificial Intelligence · Inclusive Design .

## 1 Introduction

The evolution of wearable technology is increasingly supporting the diagnosis and monitoring of neurodegenerative diseases such as dementia [1]. Dementia causes progressive cognitive decline and early dependence on caregivers, who often face significant physical and emotional challenges [2]. With an aging global population, the prevalence of dementia continues to rise. Therefore, monitoring systems that provide real-time data can help caregivers deliver more responsive and personalized care [3]. The project "Platform for Routine Measurement and Monitoring of Activity in Elderly People with Dementia" aims to develop and validate a new system for monitoring location and daily activities among elderly individuals with dementia, combining technological innovation with a user-centered design approach. This paper is organized into four sections. Section 2 "Methodology", outlines the approach used to achieve the stated goals, Section 3 "Development" presents the progress made in hardware, software, artificial intelligence (AI), and design. Finally, Section 4 "Future Work" describes the next steps planned to advance the system based on the defined methodologies.

## 2 Related work

Recent studies have demonstrated the increasing relevance of wearable technologies in enabling non-invasive, real-time monitoring of vulnerable populations, particularly those at risk for neurodegenerative diseases. For example, accelerometer-based devices have been used to assess the impact of moderate physical activity on cognitive performance in individuals with Parkinson's disease, revealing improvements in visuospatial skills, memory, and executive function [4]. Additionally, multi-modal wearable systems have been developed to detect behavioral symptoms of dementia, such as agitation and aggression, with personalized sensor configurations significantly enhancing model performance for activity classification [5]. A recent systematic review titled "Advancing Remote Monitoring for Patients With Alzheimer Disease and Related Dementias" offers a comprehensive evaluation of remote monitoring technologies for dementia care. The review highlights promising developments in wearable devices, home-based sensors, and telehealth platforms, but also emphasizes key challenges such as digital literacy barriers, sensory impairments, and technological accessibility. The authors show the importance of adapting user interfaces and involving caregivers and clinicians early in the design process to ensure adoption, usability, and equity [6].

## 3 Methodology

As part of a wider system represented in Figure 1 under development, two wearable devices are being developed: an In-ear Wearable Device (HowMI) [7], and a Wrist Wearable Device [8]. This paper focuses on the Wrist Device, which has functional requirements aimed at continuously monitoring heart rate, blood oxygen saturation, step count, sleep patterns, real-time location, falls,

and emergency situations. Its development is also guided by four domains of non-functional requirements: physical, emotional, cognitive, and implementation success. Various sensors are used to obtain these monitoring parameters, such as the BMI270 for motion tracking [9], the MAX30101 for heart rate and blood oxygen saturation measurements [10], and the SAM-M10Q for outdoor geolocation via Global Navigation Satellite System (GNSS) [11]. Indoor localization is done through Bluetooth Low Energy (BLE) communication with devices previously placed in known rooms of a building. Data acquisition is managed by embedded software running on a microcontroller, with real-time encrypted data transmission enabled via Bluetooth Low Energy (BLE). The collected data are sent to a nearby base station and then transmitted to a central AI-based processing platform, where machine learning algorithms analyze the data to recognize activity patterns and detect anomalies. Processed information is made available through a secure web portal, allowing caregivers and healthcare professionals to monitor users remotely and in real time. Figure 1 illustrates the intended operation of the system, showing the flow of information from the wearable devices to the central AI engine and the interface used by caregivers. The system is designed for deployment in diverse care environments, including senior residences and private homes.

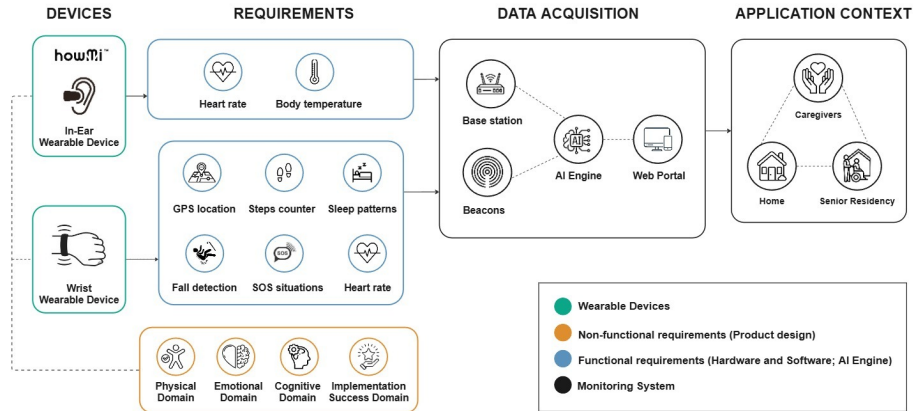


Fig. 1. Overview of the project system

## 4 Development

This section outlines ongoing improvements to a wearable device designed for monitoring elderly people with dementia. The development focuses on refining both hardware and software to enhance system performance, user comfort, and

sensor integration. Key updates include ergonomic redesigns to improve wearability, expanded data acquisition to support more robust AI model training, upgraded communication protocols for secure and efficient data transmission, and optimized power management to extend device autonomy.

#### 4.1 Hardware and Software

A functional prototype of the wearable device was developed using development boards for the selected sensors, organized into three main modules. Each sensor was individually programmed and tested: the BM270 [9] for recording steps, distance, and inertial data; the SAM-M10Q [11] for acquiring geolocation outdoor and time information; and the MAX30101 [10] for high accuracy heart rate and blood oxygen saturation measurement. Indoor location was tested by sending and receiving location-relevant information via BLE protocol. A real-time operating system (RTOS) was used to manage sequential data acquisition, and the BLE protocol was implemented and evaluated for encrypted wireless transmission to external devices.

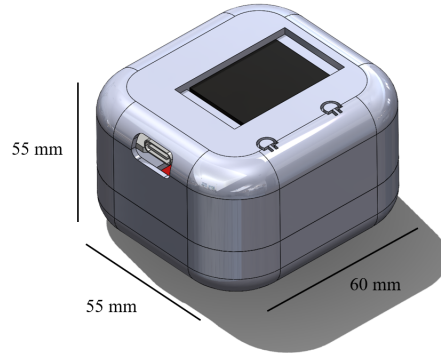
#### 4.2 Usability and Ergonomic Design

Several 3D models and test prototypes were developed to optimize the sensor layout and overall device dimensions, addressing constraints such as mechanical tolerances and space allocation for connectors. After approximately thirty iterations, a functional model was achieved, as shown in Figure 2. A square or rectangular form factor was initially selected, as it proved more suitable in terms of component arrangement and usability, although special attention was required to ensure comfort during daily use. A smaller screen size was also chosen to match the anatomy of the adult wrist, contributing to a more compact and comfortable device. Final validation will be carried out through usability testing with elderly and their caregivers to gather more accurate and practical feedback.

#### 4.3 Activities Recognition

AI techniques were developed to support real-time activity recognition and anomaly detection [12]. Machine learning models, such as Support Vector Machines (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Logistic Regression, Gradient Boosting (XGBoost, LightGBM), were trained on publicly available datasets of wearable sensors to classify activities including walking, sleeping, eating, and hygiene-related tasks. A total of 30 datasets were analyzed, varying in sensor types, activity coverage, participant demographics, and recording duration. Most datasets featured healthy adults, with only a few including elderly individuals or long-term recordings.

Deep learning, including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU), were also explored



**Fig. 2.** 3D representation of the final functional model with general dimensions

to improve the recognition of more complex behaviors. Training was conducted using an 80/10/10 split for training, validation, and testing and standard metrics including precision, recall, F1-score, accuracy, and confusion matrices were used for performance evaluation, with macro and weighted-averages reported to account for class imbalance.

Tree-based models, namely XGBoost and LightGBM performed well for common activities and imbalanced datasets, achieving average F1-scores of 0.91 and 0.90, respectively. Deep learning models, particularly GRUs, showed superior accuracy in capturing temporal dependencies and recognizing transitions between complex activities, reaching an F1-score of 0.94 and the highest recall among all models tested.

In addition to activity recognition, this study also explored the application of unsupervised learning approaches to detect deviations from users' typical daily routines. These methods do not rely on labeled anomaly data, making them particularly suitable for healthcare contexts where anomalous events, such as changes in sleep patterns, skipped meals, or irregular mobility, are rare, variable, and difficult to annotate.

Models were trained exclusively on synthetic data representing normal behavior, allowing them to learn baseline routine patterns. Subsequently, deviations were identified based on reconstruction errors or prediction discrepancies, which may reflect early signs of health deterioration, such as increased inactivity, agitation, or disrupted sleep. These AI capabilities enhance the system's ability to monitor daily activities continuously and to provide early alerts to caregivers and healthcare professionals.

Two main strategies were evaluated. The first was a GRU-based autoencoder where anomalies were detected based on reconstruction error, with a threshold defined at the 95th percentile of errors on the training set. A day was flagged as anomalous if more than 10% of its time steps were incorrectly reconstructed. This model achieved an overall accuracy of 92%, a recall of 99% for anomalies, but

lower precision (39%), due to a significant number of false positives, especially when the proportion of anomalous intervals per day was low (5%).

The second approach used a GRU-based sequence-to-sequence model with temporal embeddings and an attention mechanism to predict the next activity in a 30-step sequence. Anomalies were identified as mismatches between predicted and actual activity codes, using an error threshold defined as the mean plus two standard deviations of training errors. This method achieved near-perfect precision (99%) in identifying normal routines and high anomaly recall, but also suffered from false positives when anomaly presence was subtle.

While the results are promising, one key limitation remains, the lack of real-world datasets involving individuals with dementia.

## 5 Future Work

The work next step involves analyzing the device’s current model to align its technology and design, considering wrist anthropometry to create a final model suitable for the elderly. Key elements like buttons and straps will be studied for usability, with textures and colors differentiating button functions, and a safety system added to the strap to prevent accidental removal. Development will also focus on creating an extensive dataset for monitoring dementia, integrating advanced AI techniques to improve real-time monitoring. Data will be structured and stored via BLE on smartphones. Efforts will prioritize optimizing data acquisition algorithms, reducing power consumption, and enhancing communication protocols, including testing LoRa (Long Range) with AES-CTR (Advanced Encryption Standard - Counter) encryption for security. A hardware redesign will integrate sensors on a compact PCB (Printed Circuit Board). Software improvements will add features, optimize data processing, and enhance energy efficiency, aiming to deliver a cost-effective, user-beneficial product. Although the project is still in development stages, future work will also include usability testing with real users to assess interaction and integration into daily routines. Ethical and regulatory aspects such as informed consent, data privacy, and compliance with relevant standards will also be addressed in collaboration with partner institutions.

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